

Lecture 4: Bode Plot

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
frequency response	frekvenssvar
Bode plot	Bode plot
Bode magnitude plot	Amplitudkurva
break points	brytpunkter
step response	stegsvar
rise time	stigtiden
settling time	insvängningstiden
overshoot	överslängen
relative damping	relativa dämpning
stationary error	stationärt fel
disturbance rejection	störningsundertryckningar
gain	förstärkning
phase shift	fasvridning
non-minimum phase system	icke-minimumfastsystem
time delay	tidsfördröjning

In the previous lecture..

- ▶ Root locus
- ▶ Nyquist stability criterion

Today:

- ▶ Specifications
- ▶ Frequency response
- ▶ Bode plot



Outline

Recap

- Root Locus

- Nyquist stability criterion

Specifications

- Objective performance metrics

Frequency response

- Frequency response

- Recap: Nyquist diagram

Bode Plot: Draw $G(i\omega)$ as a function of ω

Bonus

- Non-minimum phase system

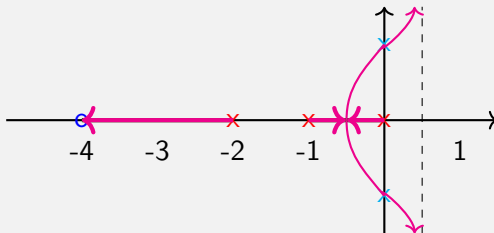
Recap: Root Locus: Step-by-step

0. (**#roots/branches**). Since $(*)$ has degree n , there are n roots for each value of K
 $\Rightarrow$ we say that the root locus has **n branches**
 Note: roots are either real or complex conjugate $\Rightarrow$ **root locus is symmetric w.r.t. Re-axis**
- I. (**starting points**). When $K = 0$, $(*)$ becomes $P(s) = 0 \Rightarrow$ The n branches of the root locus start at the roots of $P(s) \Rightarrow$ **starting points = zeros of $P(s)$**
- II. (**ending points**). Of these n branches, m will end (for $K = \infty$) at the roots of $Q(s)$
 $\Rightarrow$ **ending points = zeros of $Q(s)$**
- III. (**asymptotes**). If $m < n$, the remaining $n - m$ branches will go towards $\infty \Rightarrow$ the curves the root locus approaches as $s \rightarrow \infty$ are called **asymptotes** [Resultat 3.1]
 - For $s \rightarrow \infty$, a good approximation of $(*)$ is: $(s + \frac{a_1 - b_1}{n - m})^{n - m} = -K$
 - **direction** $\Rightarrow \arg(s + \frac{a_1 - b_1}{n - m}) = \frac{\pi}{n - m} + 2k \frac{\pi}{n - m}, k = 0, 1, 2, \dots$
 - **intersection points** $\Rightarrow -\frac{a_1 - b_1}{n - m}$
- IV. (**roots on Re-axis?**). Root locus is found on a given section of the Re-axis only if the **total # of poles and zeros of G_o to the right of the section is odd** [Resultat 3.2]
- V. (**roots on Im-axis?**). Assume $s = i\omega$ and **solve** $(*)$ w.r.t. ω and K . [Resultat 3.2]

Recap: Root Locus: Example

Example

Conclusion



Stable for $K < 6$. Oscillatory when K approaches 6. For large K , the poles are given by the zero points of the open-loop system and relative degrees (zero points at infinity).

Final observations:

K is often the gain in a P controller. However, the method is more general.

An important tool for getting an idea of what happens when you change a parameter.

Recap: The Nyquist stability criterion

$$G_c = \frac{G_o}{1 + G_o}$$

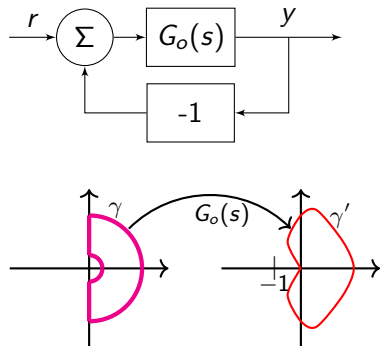
$\Rightarrow$ poles of G_c given by zeros of $1 + G_o(s)$!

Theorem (Nyquist stability criterion)

$\#$ of unstable poles of $G_o(s)$
 $=$ $\#$ of unstable poles of $G_c(s)$
 $+ \#$ of anticlockwise encirclements of -1
 made by the Nyquist plot of $G_o(s)$

Theorem (Simplified Nyquist stability criterion)

Assume that $G_o(s)$ is stable (poles in $s = 0$ OK).
 Then $G_c(s)$ is asymptotically stable
 if γ' does not encircle -1 .





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Bonus

Non-minimum phase system

Example (Step response)



Objective performance metrics

This can be measured using, among other things:

- **Rise time:** $T_r = t_2 - t_1$ where

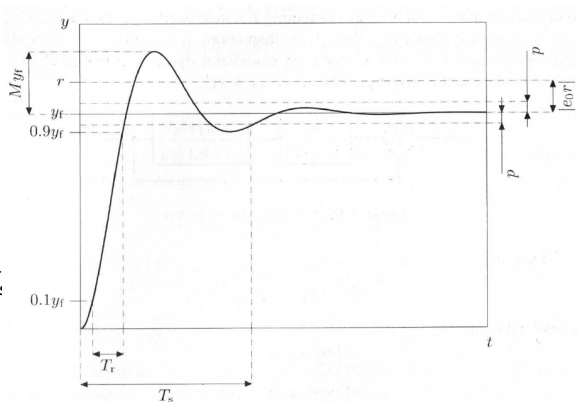
$$\begin{cases} y(t_2) = 0.9y_f \\ y(t_1) = 0.1y_f \end{cases}$$

- **Settling time:** $T_s(p\%)$, defined

$$y_f(1-p) \leq y(t) \leq y_f(1+p), \quad \forall t \geq T_s$$

(usual p values are 2% or 5%)

- **Overshoot:** $M = \frac{y_{\max} - y_f}{y_f} \cdot 100\%$

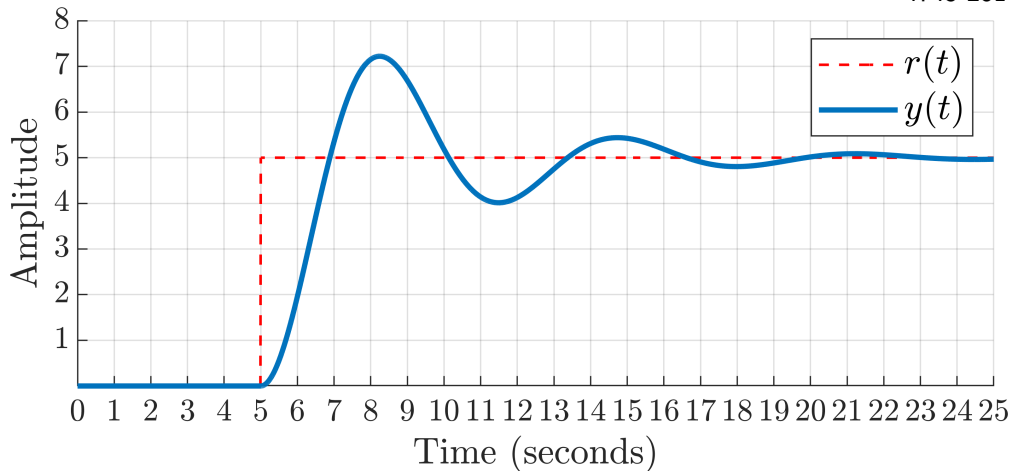


Homework: Study Fig. 3.17. Note that in this case, $y_f = 1$!

Determine $T_s(5\%)$, T_r , and M



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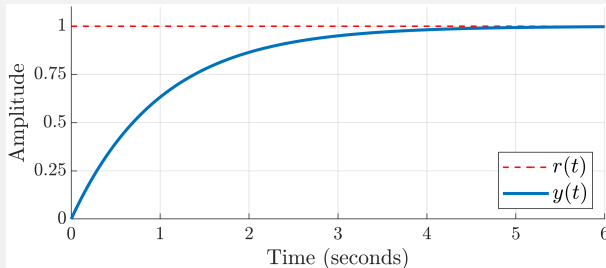
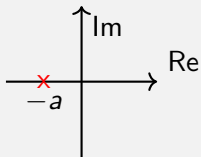


Objective performance measures: Example

Example (1st order)

Assume that $G_c(s) = \frac{a}{s+a}$.

The plots illustrate the step response (for $a = 1$) and the pole position



$$\Rightarrow T_r = \frac{2.2}{a}, \quad T_s(5\%) \approx \frac{3}{a}, \quad M = 0$$

Objective performance measures: Example

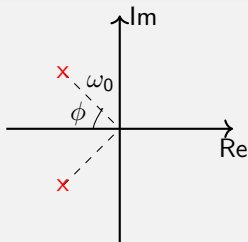
Example (2nd order)

Assume that $G_c(s) = \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$.

The parameter ξ is < 1 for complex conjugate poles and is called *relative damping*:

$$\text{poles in } s = \omega_0(\xi \pm i\sqrt{1 - \xi^2})$$

where the angle ϕ in the figure is s.t. $\xi \equiv \cos(\phi)$



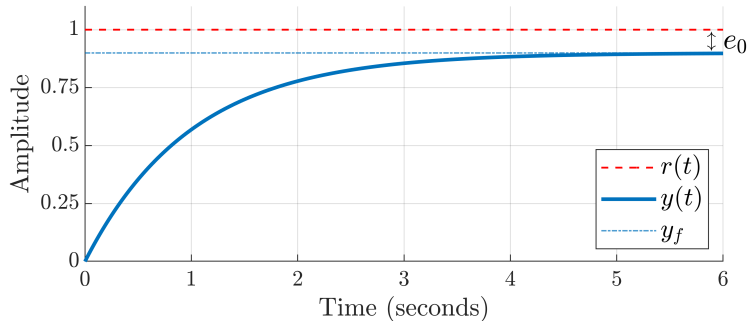
$$\Rightarrow T_r = \frac{e^{\frac{\phi}{\tan \phi}}}{\omega_0}, \quad T_s(5\%) \approx \frac{3}{\omega_0 \xi} \text{ (for small } \xi), \quad M = e^{-\frac{\pi \xi}{\sqrt{1 - \xi^2}}}$$

Objective performance measures

1. The ability of the output signal to follow a given reference signal

- $T_r, T_s(p\%), M$
- *Obs*: Important in root locus analysis/design $\Leftrightarrow$ Dominant poles

2. Stationary error



3. Other measures, such as disturbance rejection (the regulator problem)



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Frequency response

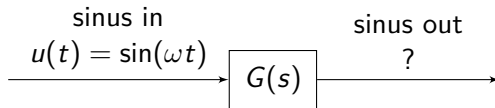
Recap: Nyquist diagram

Bode Plot: Draw $G(i\omega)$ as a function of ω

Bonus

Non-minimum phase system

Frequency response – Stable System



$$\Rightarrow y(t) = \underbrace{|G(i\omega)|}_{\text{gain}} \cdot \sin(\omega t + \underbrace{\arg G(i\omega)}_{\text{phase shift}}) + \underbrace{\text{transient}}_{\rightarrow 0 \text{ if stable}}$$

The *frequency response* is given by $G(i\omega)$.


$$|z| = r = \sqrt{x^2 + y^2}, \quad \arg z = \phi = \arctan(y/x)$$

Input $u(t) = \sin(\omega t) = \text{Im}(e^{i\omega t})$, from $t = -\infty$

Transfer function $G(s) = \int_0^\infty g(\tau)e^{-s\tau}d\tau$

$$\begin{aligned} \text{Output } y(t) &= \int_0^\infty g(\tau) u(t-\tau) d\tau \\ &= \text{Im}\left(\int_0^\infty g(\tau) e^{i\omega(t-\tau)} d\tau\right) = \text{Im}\left(\int_0^\infty g(\tau) e^{-i\omega\tau} d\tau e^{i\omega t}\right) \\ &= \text{Im}(G(i\omega) e^{i\omega t}) = \text{Im}(|G(i\omega)| e^{i \arg G(i\omega) + i\omega t}) \\ &= |G(i\omega)| \sin(\omega t + \arg G(i\omega)) \end{aligned}$$

Assume that $G(s) = s$ and that the input is $u(t) = \sin(\omega t)$.

$$\Rightarrow y(t) = \dot{u}(t) = \omega \cos(\omega t) = \omega \sin(\omega t + \frac{\pi}{2})$$

Phase shift: $+90^\circ$

Amplifies: High frequencies more than low frequencies (by a factor ω)

Assume that $G(s) = \frac{1}{s}$ and that the input is $u(t) = \sin(\omega t)$.

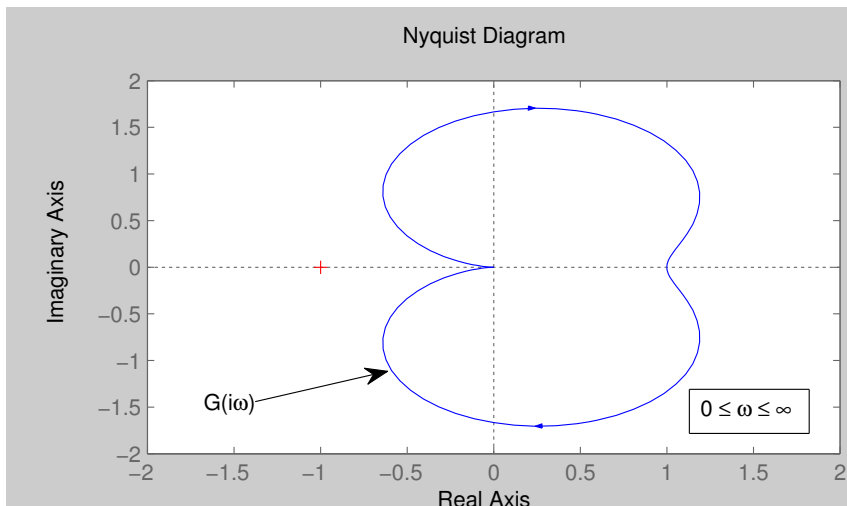
$$\Rightarrow y(t) = \int_0^t \sin(\omega\tau) d\tau = -\frac{1}{\omega} \cos(\omega t) + \text{constant} = \frac{1}{\omega} \sin\left(\omega t - \frac{\pi}{2}\right) + \text{constant}$$

Phase shift: -90°

Gain: Low frequencies more than high frequencies (by a factor $\frac{1}{\omega}$)

Draw $G(i\omega)$ as a function of ω

Nyquist diagram:



Difficult to read frequency!

Recap: Nyquist diagram

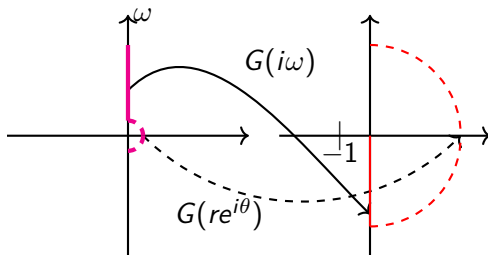
Example

$$G(s) = \frac{1}{s}, \quad \Rightarrow \quad G(i\omega) = \frac{1}{i\omega} = \frac{-i}{\omega}, \quad r \leq \omega < R$$

What happens on the small circle around the origin:

$$s = re^{i\theta}, \quad -\pi/2 \leq \theta \leq \pi/2 \quad \Rightarrow \quad G(re^{i\theta}) = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}$$

It does not circle -1 , i.e., stable with feedback!



Recap: Nyquist diagram

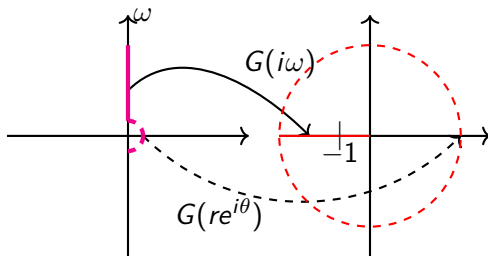
Example

$$G(s) = \frac{1}{s^2}, \quad \Rightarrow \quad G(i\omega) = \frac{1}{(i\omega)^2} = \frac{-1}{\omega^2}, \quad r \leq \omega < R$$

What happens on the small circle around the origin:

$$s = re^{i\theta}, \quad -\pi/2 \leq \theta \leq \pi/2 \quad \Rightarrow \quad G(re^{i\theta}) = \frac{1}{r^2 e^{i2\theta}} = \frac{1}{r^2} e^{-i2\theta}$$

Passes through -1 , i.e., not asymptotically stable with feedback!





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Bonus

- Non-minimum phase system

Note:

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Assume that

then the frequency response is given by

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Bode magnitude plot

Example (fort.)

Bode magnitude plot:

$$\log |G(i\omega)| = \log K + \log \sqrt{1 + \left(\frac{\omega}{b}\right)^2} - \log \omega - \log \sqrt{1 + \left(\frac{\omega}{a}\right)^2}$$

Approximation:

$$\log \sqrt{1 + \left(\frac{\omega}{b}\right)^2} = \frac{1}{2} \log \left(1 + \left(\frac{\omega}{b}\right)^2\right) \approx \begin{cases} 0, & \omega < b, \quad \omega \text{ small} \\ \log \omega - \log b, & \omega \geq b, \quad \omega \text{ big} \end{cases}$$

- ▶ The slope increases by $+1 \approx 20$ dB/decade.
- ▶ Maximum error at $\omega = b \Rightarrow \frac{1}{2} \log 2 = 0.15 = 3$ dB.



Approximate Bode magnitude plot

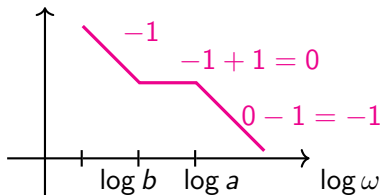
Summary:

1. We find the *break points* at the frequencies

$$\omega = b \quad \text{which gives the slope } +1$$

$$\omega = a \quad \text{which gives the slope } -1$$

2. What happens when $\omega \rightarrow 0$? (leftmost part of the x-axis, as $\log_{10}(0) = -\infty$)
If ω is small, the slope = number of integrators = -1 .



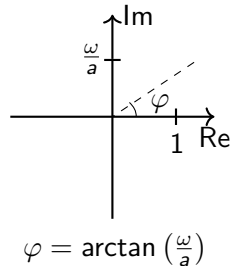
- In practice, Bode plots are drawn using a computer.
- In the exercises, it is easiest to calculate the exact values at the breakpoints and at a few more points and then interpolate.

Phase plot

It is also common to draw a phase curve in a Bode plot.

$$\arg(G(i\omega)) = \arg\left(\frac{1 + \frac{i\omega}{b}}{i\omega(1 + \frac{i\omega}{a})}\right) = \arg\left(1 + \frac{i\omega}{b}\right) - \arg(i\omega) - \arg\left(1 + \frac{i\omega}{a}\right)$$

$$\Rightarrow \arg(G(i\omega)) = \arctan\left(\frac{\omega}{b}\right) - 90^\circ - \arctan\left(\frac{\omega}{a}\right)$$



- In practice, the phase plot is also drawn using a computer.
- In the exercises, it is easiest to calculate the exact values at the breakpoints and at a few more points, and then interpolate.

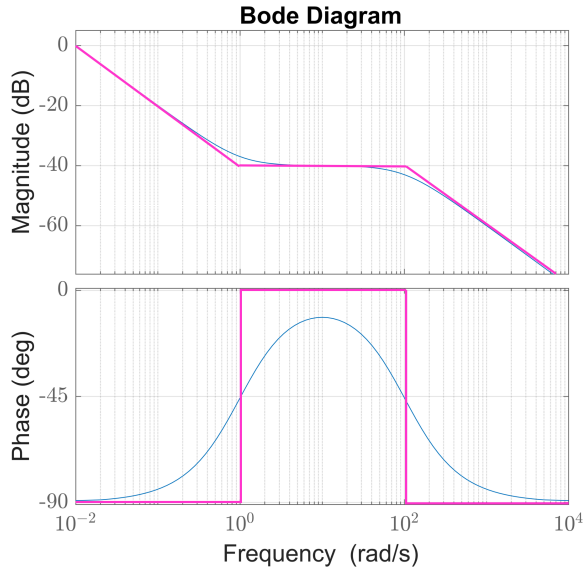
Phase plot: arctan & mnemonic

Remember that:

- ▶ $\arctan(0) = 0^\circ$
- ▶ $\arctan(1) = 45^\circ$
- ▶ $\arctan(\infty) = 90^\circ$

Mnemonic/Minnesregel (informal): If the magnitude Bode plot has a line with slope $\pm k$, then the value of the phase is $\pm k 90^\circ$

Bode Plot: Example



$$G(s) = \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$\xi = \text{relative damping}$: it takes values in the interval $0 < \xi < 1$.

Slope:

$$\begin{cases} \omega \text{ small} & 0 \\ \omega \text{ big} & -2 = -40\text{dB/decade} \\ \omega = \omega_0 & \Rightarrow \log |G(i\omega_0)| = \log \frac{1}{2\zeta} \end{cases}$$

Small damping $\Rightarrow$ Large amplification of frequencies close to ω_0 !

Draw carefully!

Homework: Read pp. 86-87 and study Fig. 4.3!

Draw (MATLAB) the Bode plot for $\omega_0 = 1 = 10^0$ and different values of ξ : What do you expect?



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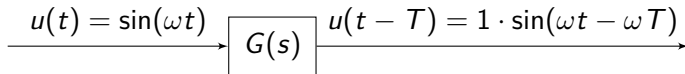
Non-minimum phase system

$$G_1(s) = \frac{1 + s/b}{1 + s/a} \quad G_2(s) = \frac{1 - s/b}{1 + s/a}$$
$$|G_1(i\omega)| = |G_2(i\omega)|$$
$$\begin{cases} \arg(G_1(i\omega)) = \arctan \frac{\omega}{b} - \arctan \frac{\omega}{a} \\ \arg(G_2(i\omega)) = -\arctan \frac{\omega}{b} - \arctan \frac{\omega}{a} \end{cases} \Rightarrow \arg(G_2(i\omega)) < \arg(G_1(i\omega))$$

$\Rightarrow G_2$ phase shifts more (negatively) than G_1 !

Minimum-phase = All the zeros and all the poles in LHP

Time delay



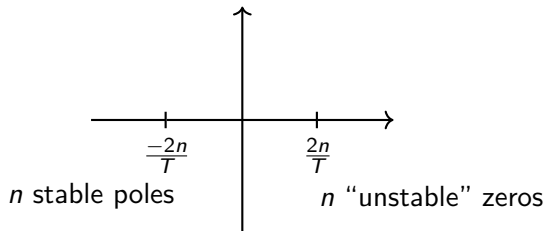
$$\Rightarrow G(i\omega) = 1 \cdot e^{-i\omega T} \Rightarrow G(s) = e^{-sT}$$

We see that $|G(i\omega)| = 1$ and that $\arg(G(i\omega)) = -\omega T$.

Minimum phase? Compute poles and zeros

$$e^{-sT} = \left(\frac{e^{-\frac{sT}{2n}}}{e^{+\frac{sT}{2n}}} \right)^n \approx \frac{\left(1 - \frac{sT}{2n}\right)^n}{\left(1 + \frac{sT}{2n}\right)^n}$$

$\Rightarrow$ Non-minimum phase!



Bode's relation

Bode's relation: From the Bode plots, there appears to be a relation between the gain curve and the phase curve (minimum-phase systems)

Theorem (Bodes relation)

Let $G(s)$ be minimum-phase and $G(0) > 0$.

If $|G(i\omega)|$ in a certain frequency range decreases by

$$-20\text{dB/decade} \xRightarrow{\text{then}} \arg(G(i\omega)) \approx -90^\circ$$

$$-40\text{dB/decade} \xRightarrow{\text{then}} \arg(G(i\omega)) \approx -180^\circ$$

etc.

For complex numbers:

$$\log G(i\omega) = \log |G(i\omega)| + i \arg(G(i\omega))$$